

Chapter 2: Basic Structures: Sets, Functions, Sequences, Sums and Matrices

2.1 Sets

2.1.1 Introduction

Definition 1: A set is an unordered collection of distinct objects, called elements or members of the set. A set is said to contain its elements. We write $a \in A$ to denote that a is an element of the set A . The notation $a \notin A$ denotes that a is not an element of the set A .

Note: This definition is an intuitive definition, which is not part of a formal theory of sets.

- Sets are denoted using uppercase letters.
- Elements of sets are denoted by lowercase letters.

Description of a Set

- One way is to list all the members of a set, when this is possible.
- We use a notation where all members of the set are listed between braces.
- For example, the notation $\{a, b, c, d\}$ represents the set with the four elements a, b, c and d . This way of describing a set is known as the **roster method**.

Example 1:

The set V of all vowels in the English alphabet can be written as $V = \{a, e, i, o, u\}$. $\square$

Example 2:

The set O of odd positive integers less than 10 can be expressed by $O = \{1, 3, 5, 7, 9\}$. $\square$